

1a.

$$\phi = \phi_{ABCD} + \phi_{MNPQ} + \phi_{ADQM} + \phi_{BCPN} + \phi_{CDQP} + \phi_{ABHN}$$

$$\begin{aligned}\phi_{ABCD} &= \int \vec{E} \cdot (d\vec{x} d\vec{y}) (\hat{k}) \\ &= z^2 \iint dx dy = a^4\end{aligned}$$

$$\begin{aligned}\phi_{MNPQ} &= \int \vec{E} \cdot (d\vec{x} d\vec{y}) (-\hat{k}) \\ &= -z^2 \iint dx dy = 0\end{aligned}$$

$$\begin{aligned}\phi_{ADQM} &= \int \vec{E} \cdot (d\vec{x} d\vec{z}) (-\hat{j}) \\ &= -\int xy (dx dz) = 0\end{aligned}$$

$$\begin{aligned}\phi_{BCPN} &= \int \vec{E} \cdot (d\vec{x} d\vec{z}) (\hat{j}) \\ &= \int xy (dx dz) = \frac{a^4}{2}\end{aligned}$$

$$\begin{aligned}\phi_{CDQP} &= \int \vec{E} \cdot (d\vec{y} d\vec{z}) (-\hat{k}) \\ &= -(zx^2 + 1) \iint dy dz \\ &= -(a)(a) = -a^2\end{aligned}$$

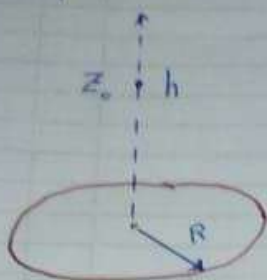
$$\begin{aligned}\phi_{ABHN} &= \int \vec{E} \cdot (d\vec{y} d\vec{z}) \hat{j} \\ &= (zx^2 + 1) \iint dy dz \\ &= (za^2 + 1)(a^2)\end{aligned}$$

$$\begin{aligned}\Rightarrow \phi &= a^4 + 0 + 0 + \frac{a^4}{2} - a^2 + (za^2 + 1)a^2 \\ \phi &= \frac{7}{2} a^4\end{aligned}$$

$$1b. \phi = \frac{q_e}{\epsilon_0}$$

$$\rightarrow q_e = \frac{z}{2} a^4 \epsilon_0$$

2. $\vec{E}$ para un anillo de carga uniforme λ_0



$$\vec{E}_z = \frac{\lambda_0 R z \hat{k}}{2 \epsilon_0 (R^2 + z^2)^{3/2}}$$

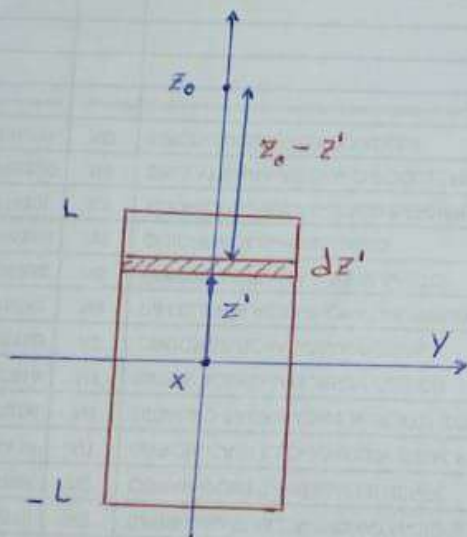
$$d\lambda_0 = \sigma_0 dz'$$

$\vec{E}$ para un cilindro hueco sin tapas de carga uniforme σ_0
Haciendo los cambios para un dz' en el cilindro:

$$E_z \rightarrow dE_z$$

$$\lambda_0 \rightarrow d\lambda_0$$

$$z \rightarrow (z_0 - z')$$



$$\Rightarrow dE_z = \frac{\sigma_0 R (z_0 - z') dz'}{2 \epsilon_0 [R^2 + (z_0 - z')^2]^{3/2}}$$

$$E_z = \frac{\sigma_0 R}{2 \epsilon_0} \int_{-L}^L \frac{(z_0 - z') dz'}{[R^2 + (z_0 - z')^2]^{3/2}}$$

$$\Rightarrow V(z) = -\frac{\sigma_0 R}{2 \epsilon_0} \ln \left(\frac{\sqrt{R^2 + (z - \frac{h}{2})^2} + (z - \frac{h}{2})}{\sqrt{R^2 + (z + \frac{h}{2})^2} + (z + \frac{h}{2})} \right)$$

Problema 3 (5 puntos)

Una distribución de carga no uniforme pero esféricamente simétrica, de radio $2R$, tiene una densidad de carga ρ dada como sigue:

$$\rho(r) = \begin{cases} \frac{\rho_0}{2} & \text{para } r \leq R \\ \frac{\rho_0}{2} \left(1 - \frac{r}{2R}\right) & \text{para } R < r \leq 2R \\ 0 & \text{para } 2R < r \end{cases}$$

Determinar el campo eléctrico en todo el espacio.

1. $r \leq R$

$$\frac{dQ}{dV} = \rho = \frac{\rho_0}{2}$$

$$\int_0^R dQ = \frac{\rho_0}{2} \int_0^R (4\pi r^2 dr)$$

$$Q_e = \frac{2}{3} \pi \rho_0 R^3 \quad (\text{carga encerrada})$$

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_e}{\epsilon_0}$$

$$\rightarrow \vec{E} = \frac{\rho_0}{6\epsilon_0} r \hat{e}_r$$

$R < r \leq 2R$

$$Q_e = \frac{2}{3} \pi \rho_0 R^3 + Q_R$$

$$\frac{dQ}{dV} = \rho_0 \left(1 - \frac{r}{2R}\right)$$

$$\int_{Q_R}^{Q_r} dQ = \int_R^r \rho_0 \left(1 - \frac{r}{2R}\right) (4\pi r^2 dr)$$

$$Q_r - Q_R = 2\pi \rho_0 \left(\frac{r^3}{3} - \frac{r^4}{8R} - \frac{5R^3}{24} \right)$$

$$Q_r = 2\pi \rho_0 \left(\frac{r^3}{3} - \frac{r^4}{8R} - \frac{5R^3}{24} \right) + Q_R \quad \dots (1)$$

Es la nueva carga encerrada.

$$\rightarrow \oint \vec{E} \cdot d\vec{S} = \frac{Q_r}{\epsilon_0}$$

$$\vec{E} = \frac{\rho_0}{2\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{8R} + \frac{R^3}{8r^2} \right) \hat{e}_r$$

Para $r > 2R$

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{2R}}{\epsilon_0}$$

$$Q_{2R} = Q_R + Q_{R \rightarrow 2R}$$

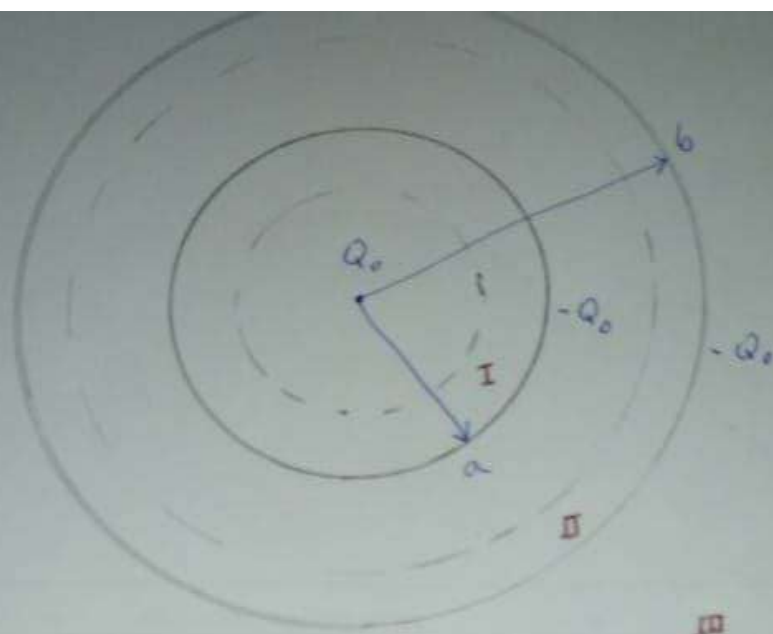
De (1):

$$Q_{2R} = \frac{19}{12} \pi \rho_0 R^3$$

$$\Rightarrow E (4\pi r^2) = \frac{19}{12} \frac{\pi \rho_0 R^3}{\epsilon_0}$$

Finalmente:

$$\vec{E} = \frac{19}{48} \frac{R^3 \rho_0}{r^2 \epsilon_0} \hat{e}_r$$



a)

$$E(r) = \begin{cases} \frac{kQ_0}{r^2} & \text{para } r < a \\ 0 & \text{para } a < r < b \\ -\frac{kQ_0}{r^2} & \text{para } r > b \end{cases}$$

b)

$$V(r) = \begin{cases} \frac{kQ_0}{r} + C_a = V & r < a \\ \frac{kQ_0}{a} + C_a = V_{\text{int}} & a < r < b \\ -\frac{kQ_0}{r} + C_b = V & r > b \end{cases}$$

$$\Rightarrow V_{\infty} = 0 \Rightarrow \underline{C_b = 0} \wedge V = -\frac{kQ_0}{r} \quad \underline{r > b}$$

$$r = b \Rightarrow \underline{V_b = -\frac{kQ_0}{b}} \quad \underline{a < r < b}$$

$$\therefore \frac{kQ_0}{a} + C_a = -\frac{kQ_0}{b} \Rightarrow \underline{C_a = -kQ_0 \left(\frac{1}{a} + \frac{1}{b} \right)}$$

$$\wedge V = \frac{kQ_0}{r} - kQ_0 \left(\frac{1}{a} + \frac{1}{b} \right) \rightarrow \underline{0 < r < a}$$

$$V(a) = \frac{kQ_0}{a} - kQ_0 \left(\frac{1}{a} + \frac{1}{b} \right) = -\frac{kQ_0}{b}$$